OPERATIONS PERFORMANCE REPORT

AI-Powered Issue Detection & Root Cause Analysis

Generated: April 28, 2026

Data: ops_data_simulated.csv | Weeks 1-12 | 3 Units | 4 Stages

Issues Detected: 3 CRITICAL | 3 MEDIUM

01 — EXECUTIVE SUMMARY

Warehouse A experienced a processing time surge in packaging operations starting in Week 5, jumping from 2.99 hours to 3.49 hours, which triggered a cascading failure across the entire operation. This packaging bottleneck caused delays to spike from 4.9% to 13.6% by Week 7, then further deteriorate to 19.7% by Week 9, ultimately driving customer complaints from 2.0% to 12.9% and costs from $12.29 to $14.61 in the same period. The dysfunction has now spread to Warehouse B, which shows delays at 7.1% and complaints at 4.6%, indicating systemic contamination likely from rerouted orders or shared resource constraints. Warehouse A's complaint rate of 12.9% represents the highest business impact, affecting customer retention and brand reputation at a severity score of 764.0, while the delay rate of 19.7% creates compounding operational dysfunction. Immediate intervention is required at Warehouse A's packaging stage to stop the hemorrhaging, followed by capacity rebalancing and customer damage control.

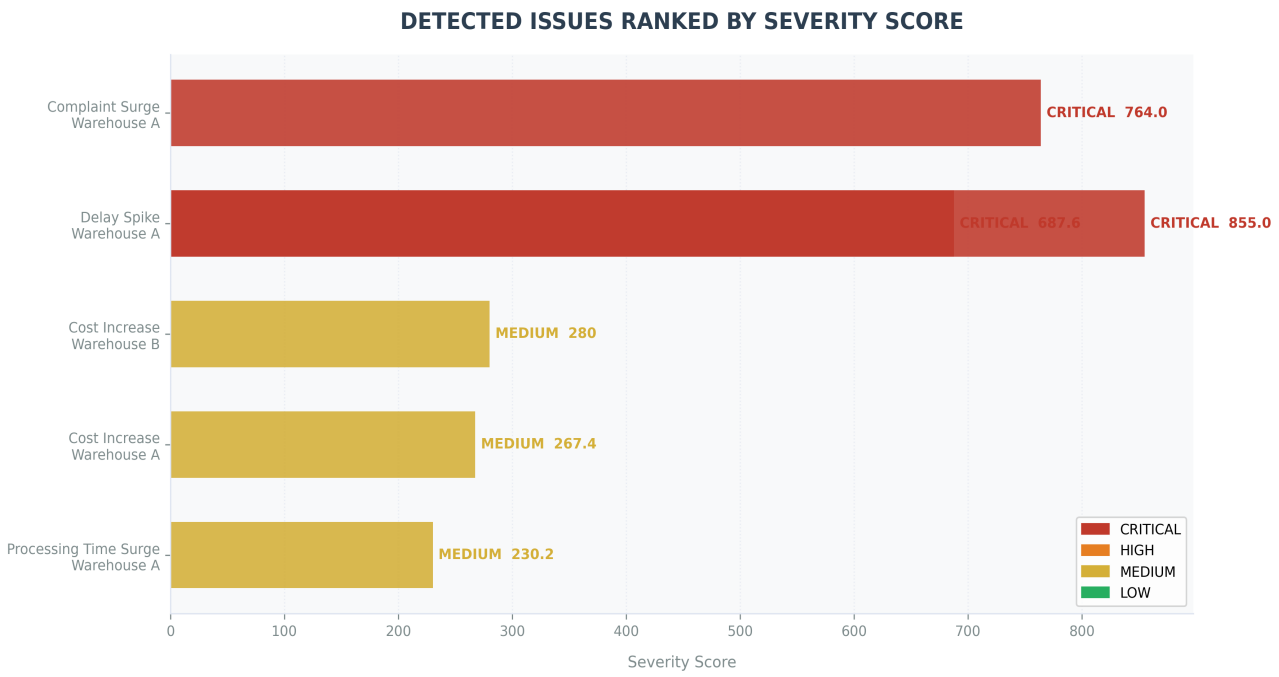
02 — BASELINE vs CURRENT PERFORMANCE

Unit	Base Delay	Curr Delay	Change	Base Complaints	Curr Complaints	Change
Support Team	4.9%	4.8%	-2.9%	2.0%	1.9%	-1.3%
Warehouse A	4.9%	12.2%	+150.4%	1.9%	7.0%	+267.2%
Warehouse B	4.9%	7.1%	+44.3%	1.9%	4.6%	+141.0%

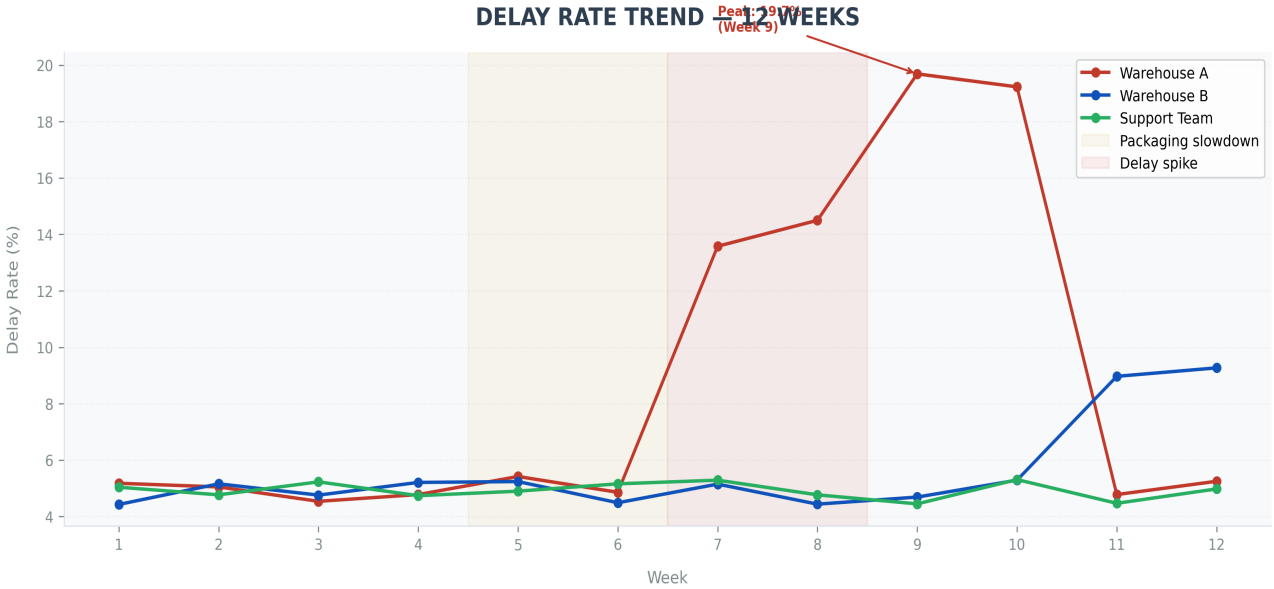
03 — TOP ISSUES RANKED BY SEVERITY

Rank	Problem	Unit	Stage	Before	After	Severity	Score
1	Delay Spike	Warehouse A	multi-stage	4.9%	13.6%	CRITICAL	855.0
2	Complaint Surge	Warehouse A	customer-facing	2.0%	12.9%	CRITICAL	764.0
3	Delay Spike	Warehouse A	multi-stage	14.5%	19.7%	CRITICAL	687.6
4	Cost Increase	Warehouse B	operations	$12.15	$13.39	MEDIUM	280
5	Cost Increase	Warehouse A	operations	$12.29	$14.61	MEDIUM	267.4
6	Processing Time Surge	Warehouse A	packaging	2.99h	3.49h	MEDIUM	230.2

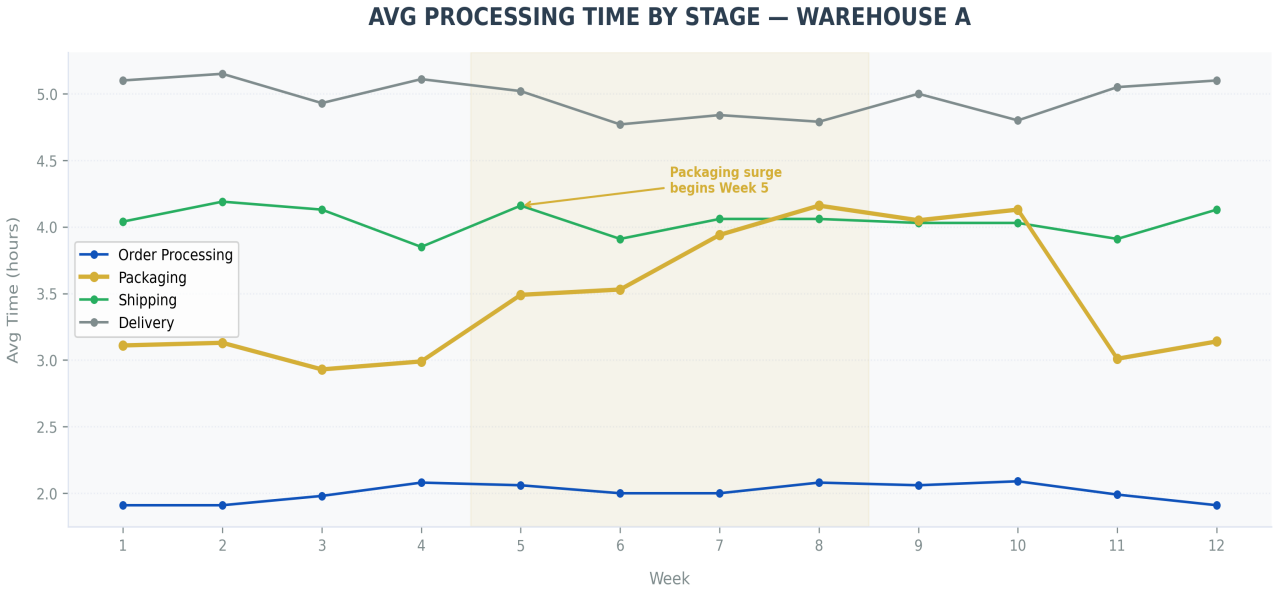
04 — SEVERITY RANKING CHART



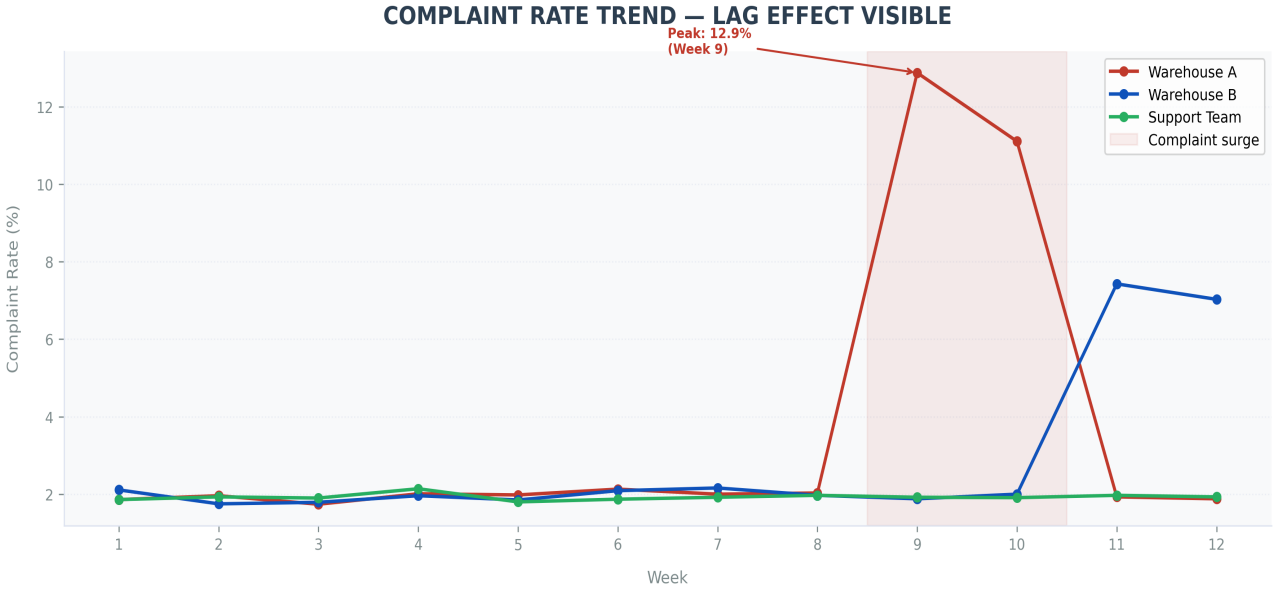
05 — DELAY RATE TREND



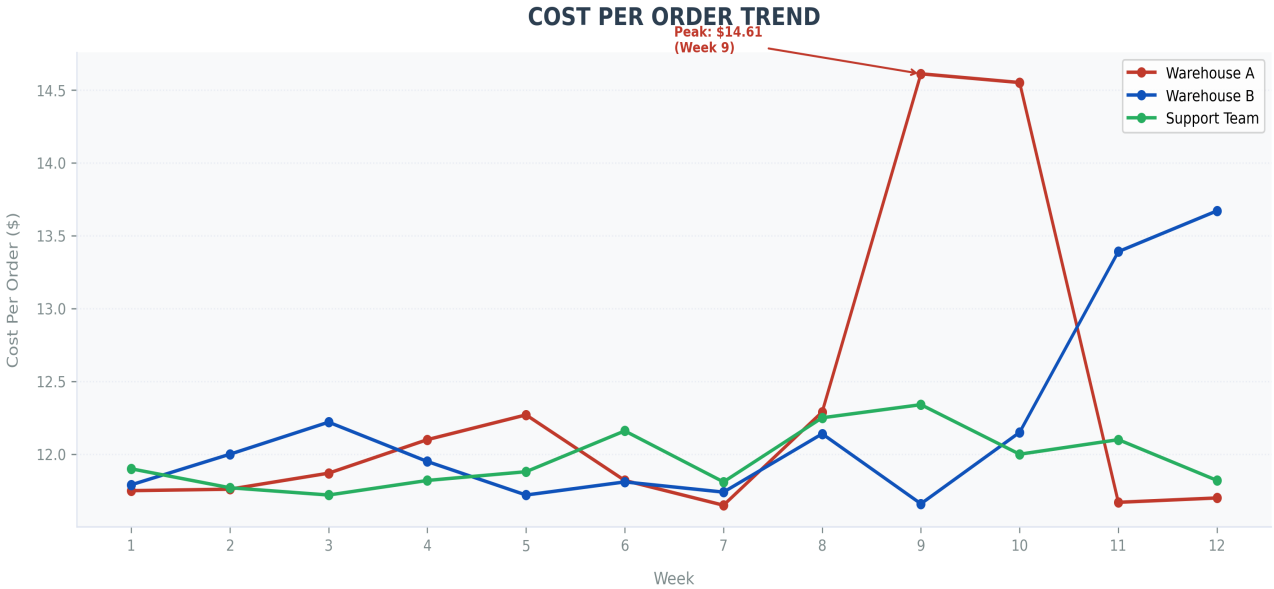
06 — PROCESSING TIME BY STAGE



07 — COMPLAINT RATE TREND



08 — COST PER ORDER TREND



09 — UNIT COMPARISON — BASELINE vs CURRENT

10 — ROOT CAUSE ANALYSIS

The root cause is the processing time surge in Warehouse A's packaging stage that first appeared in Week 5, increasing from 2.99 hours to 3.49 hours—a 16.7% deterioration. This is identified as the root cause because it precedes all other detected issues chronologically and operates at the operational core where physical order fulfillment occurs. The packaging bottleneck created immediate capacity constraints that prevented Warehouse A from maintaining its baseline throughput rate.

The propagation occurred in four distinct waves. First, the packaging slowdown directly caused the delay spike in Week 7, where Warehouse A's multi-stage delay rate jumped from 4.9% to 13.6%, representing a 177% increase over baseline. Second, by Week 9, the sustained delays compounded into a secondary delay spike, pushing the rate from 14.5% to 19.7%—indicating that orders delayed in earlier weeks were still unresolved and accumulating. Third, also in Week 9, the customer-facing impact materialized as complaint rates surged from 2.0% to 12.9%, a 545% increase that demonstrates customers experiencing the cumulative effect of weeks of delayed orders. Fourth, the operational strain forced cost increases in Week 9, with Warehouse A's cost per order rising from $12.29 to $14.61, likely driven by expedited shipping, overtime labor, and inefficiency penalties.

The contamination reached Warehouse B by Week 11, where costs increased from $12.15 to $13.39, occurring six weeks after the original packaging failure. This cross-facility spread indicates either order rerouting from Warehouse A to Warehouse B or shared resource depletion across the network. Warehouse B's delays at 7.1% and complaints at 4.6% confirm it is absorbing overflow strain from Warehouse A's dysfunction.

11 — IMPACT ANALYSIS

The highest business impact issue is the complaint surge at Warehouse A, currently at 12.9% with a severity score of 764.0. This customer-facing metric represents direct brand damage and revenue risk, as complaint rates of this magnitude typically correlate with customer churn, negative reviews, and lost lifetime value. The 12.9% rate means approximately one in eight customers served by Warehouse A are filing complaints, creating an unsustainable service failure that will compound into market reputation damage.

The delay rate of 19.7% at Warehouse A ranks second in business impact with a severity score of 687.6. This peak delay rate indicates that nearly one in five orders is experiencing delays, which feeds directly into the complaint rate and creates operational gridlock as delayed orders consume resources needed for new orders. The progression from 4.9% baseline to 13.6% in Week 7, then to 19.7% by Week 9, shows accelerating deterioration rather than stabilization.

The cost impact, while scored as medium severity, represents $2.32 per order increase at Warehouse A ($14.61 current vs $12.29 baseline), translating to a 18.9% cost escalation. Combined with Warehouse B's $1.24 per order increase ($13.39 vs $12.15), the network is bleeding operational margin. Support Team remains stable with minimal deviation, confirming the problem is contained to warehouse operations rather than systemic across all units.

12 — RECOMMENDED ACTIONS

1. Deploy emergency packaging capacity at Warehouse A immediately: Add second shift staffing or bring in temporary packaging labor to reduce processing time from 3.49 hours back toward the 2.99-hour baseline, targeting implementation within 72 hours. This addresses the root cause directly and is prerequisite to all downstream recovery.

2. Implement order flow redistribution protocol by end of Week 13: Redirect 30-40% of Warehouse A's incoming order volume to Warehouse B or other facilities while maintaining Warehouse B's delay rate below 10%, preventing further deterioration of Warehouse A's 19.7% delay rate while packaging capacity is restored.

3. Execute proactive customer outreach for all Warehouse A orders delayed beyond 5 days: Contact affected customers with delivery updates, compensation offers, or expedited shipping to mitigate the 12.9% complaint rate and prevent further escalation, targeting completion within one week.

4. Conduct forensic analysis of Warehouse A packaging operations by Week 14: Identify whether the Week 5 processing time surge originated from equipment failure, staffing shortage, training gaps, or process changes, and implement permanent corrective measures to prevent recurrence.

5. Establish daily cross-warehouse performance monitoring with escalation triggers: Set alert thresholds at 8% delay rate and 4% complaint rate for all facilities to detect future contamination patterns before they reach critical severity, with mandatory leadership review when thresholds are breached.

13 — DECISION PRIORITY TABLE

Prevents rec

AI narrative generated by Claude (Anthropic) using structured operational data. Detection logic is rule-based with severity scoring. Baseline: Weeks 1-4. Current: Weeks 9-12.